

Package ‘micsPlusTableR’

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Title MICS Plus Excel Tabulation and Review Application

Version 0.2.7

Description Runs interactive and scriptable workflows for preparing, tabulating, checking, reviewing, and exporting MICS Plus tables from an Excel tabulation plan, survey data files, and a survey-specific preparation script.

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micsPlusTableR-package

micsPlusTableR: MICS Plus Excel tabulation and review

Description

Prepare, tabulate, check, review, and export MICS Plus survey tables using an Excel tabulation plan. Work interactively with [run_app\(\)](#) or use the scripting functions with an isolated [mics_session\(\)](#).

Inputs

Supply an Excel tabulation plan with an IDX worksheet and a worksheet for each table, household and household-member SPSS files, and a survey-specific R preparation script that reads hh_path and hl_path and creates data frames named hh and hl. A sibling FIES-inputs directory is optional.

Interactive workflow

Start [run_app\(\)](#), select the survey and wave, provide the inputs, prepare the data, calculate tables, review checks, and export the results. The app creates micsPlusTableR-output in the project root by default; use `run_app(output_dir = "path/to/output")` to select another location.

Scripting workflow

The main steps are:

- [mics_session\(\)](#) creates an isolated session.
- [prepare_mics_data\(\)](#) runs the preparation script.
- [tabulate_mics_table\(\)](#) calculates cell results.
- [check_mics_table\(\)](#) checks consistency.
- [pivot_mics_table\(\)](#) arranges results for review.
- [create_excel_workbooks\(\)](#) creates output workbooks.
- [write_mics_table\(\)](#) writes the results.
- [write_mics_footnotes\(\)](#) writes table footnotes.
- [read_previous_mics_table\(\)](#) reads an earlier output.
- [compare_mics_tables\(\)](#) compares two outputs.

See [offline-workflow](#) for a complete scripting example.

Survey and wave configuration

Maintain inst/extdata/survey_choices.csv in the package source. It remains a plain, editable CSV on GitHub and is bundled automatically on installation. See [survey_choices](#) for its columns, validation rules, and update workflow.

Outputs and interpretation

Outputs include Excel workbooks and long-format observations. The app offers Excel, CSV, and RDS downloads of the full prepared long-format dataset. Use `as_sdmx_compatible()` to convert observations to the project's SDMX-compatible contract. Use `validate_sdmx_compatible()` to check them. This contract does not by itself establish compliance with an external SDMX data structure. Table checks assess consistency; analysts remain responsible for preparation scripts, survey design, denominators, and interpretation of estimates.

Author(s)

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See Also

[run_app\(\)](#), [survey_choices](#), [offline-workflow](#)

Examples

```
session <- mics_session()
session
system.file("extdata", "survey_choices.csv", package = "micsPlusTableR")
```

as_sdmx_compatible	<i>Add an SDMX-compatible observation profile to MICS cell results</i>
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Description

Converts either the package's engine cell results or an already standardized MICS cell table into a 35-variable long-format contract adapted from the mics-translation project. The output omits the custom `table_id`, `value`, `estimate`, `digits`, and `unw` duplicates; use `TABLE_ID`, `OBS_VALUE`, `DECIMALS`, and `denominator_n` instead. Input fields are unchanged. Source coordinates remain in `row_order` and `column_order`; the redundant `row_id`, `column_id`, `ROW_ID`, and `COLUMN_ID` fields are omitted. The output also omits `unweighted_n`, which duplicates `denominator_n` for engine cells. For standardized input, the supplied `denominator_n` is retained. The optional input `unit_measure` override is applied to `UNIT_MEASURE` but is not included as a separate output field.

Usage

```
as_sdmx_compatible(cells, config, table_id = NULL, table_context = NULL)
```

Arguments

<code>cells</code>	A data frame with one row per table cell.
<code>config</code>	A list containing frequency, ref_area, survey_id, and time_period.
<code>table_id</code>	Table identifier. Required for raw engine cell results.
<code>table_context</code>	Optional parsed tabulation context used to recover multi-level column headers and suppression settings.

Value

A tibble with one row per observation and the 35 standardized fields.

check_mics_table	<i>Run all applicable consistency checks</i>
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Description

Run all applicable consistency checks

Usage

```
check_mics_table(session, table = NULL, tolerance = 1e-06)
```

Arguments

session	A tabulation session.
table	Cell results; defaults to the session's latest result.
tolerance	Maximum absolute difference treated as consistent.

Value

A list containing a summary data frame and detailed check tables.

compare_mics_tables	<i>Compare two generated tables</i>
---------------------	-------------------------------------

Description

Compare two generated tables

Usage

```
compare_mics_tables(
  previous,
  current,
  ignore_col_names = FALSE,
  show_diff_over = 0
)
```

Arguments

previous	Previous or approved table.
current	Current table.
ignore_col_names	Compare columns by position instead of name.
show_diff_over	Minimum absolute numeric difference to report.

Value

A data frame containing differing cells.

create_excel_workbooks

Create output workbooks from a tabulation plan

Description

Copies the tabulation plan to standard output names without overwriting an existing file. A numeric suffix is added after a filename collision.

Usage

```
create_excel_workbooks(
  path_tab_excel,
  output_dir,
  country_code,
  period,
  wave,
  target = c("both", "output", "formatted"),
  time_tag = format(Sys.time(), "%Y%m%d%H%M%S")
)
```

Arguments

path_tab_excel	Existing .xls or .xlsx tabulation plan.
output_dir	Directory where workbooks will be created.
country_code	Short country code used in filenames.
period	Survey period used in filenames.
wave	Wave label used in filenames.
target	One of "output", "formatted", or "both".
time_tag	Timestamp used in filenames.

Value

A list with output and formatted paths.

mics_session

Create a non-Shiny MICS Plus tabulation session

Description

Create a non-Shiny MICS Plus tabulation session

Usage

```
mics_session(hh = NULL, hl = NULL)
```

Arguments

- hh Optional prepared household data frame.
- hl Optional prepared household-member data frame.

Value

A mutable, isolated tabulation session environment.

mics_table_name	<i>Get the current table title</i>
-----------------	------------------------------------

Description

Get the current table title

Usage

```
mics_table_name(session)
```

Arguments

- session A session with a tabulation sheet already read.

Value

The table title as a character scalar.

offline-workflow	<i>Run MICS Plus tabulation workflows outside Shiny</i>
------------------	---

Description

These functions provide the supported non-Shiny interface to the packaged tabulation engine. The explicit session keeps survey data, plan context, extra tables, and calculated results together without writing them into the user’s global workspace.

Details

Prepare survey data in a new session, calculate a table from the Excel plan, check its consistency, and pivot it for review before writing results. The example requires analyst-supplied files. See the package overview for links to the function references and workbook export steps.

See Also

[micsPlusTableR-package](#)

Examples

```
## Not run:
session <- mics_session()
prepare_mics_data(
  session,
  "hh.sav",
  "hl.sav",
  prep_script = "survey-preparation/survey_prep.R",
  output_dir = "micsPlusTableR-output"
)
result <- tabulate_mics_table(session, "TabulationPlan.xlsx", "1.1")
checks <- check_mics_table(session, result)
preview <- pivot_mics_table(session, result, type = "header")

## End(Not run)
```

<code>pivot_mics_table</code>	<i>Pivot long-format cell results for review</i>
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Description

Pivot long-format cell results for review

Usage

```
pivot_mics_table(
  session,
  table = NULL,
  formatted = FALSE,
  type = c("index", "header", "logic")
)
```

Arguments

<code>session</code>	A tabulation session.
<code>table</code>	Cell results; defaults to the session's latest result.
<code>formatted</code>	FALSE, TRUE, or "view" as supported by the engine.
<code>type</code>	One of "index", "header", or "logic".

Value

A wide review table.

prepare_mics_data	<i>Run a survey preparation script outside the Shiny application</i>
-------------------	--

Description

Run a survey preparation script outside the Shiny application

Usage

```
prepare_mics_data(session, hh_path, hl_path, prep_script, output_dir = NULL)
```

Arguments

session	A session created by <code>mics_session()</code> .
hh_path	Path to the household .sav file.
hl_path	Path to the household-member .sav file.
prep_script	Path to the trusted preparation .R script. A sibling FIES-inputs folder is detected automatically.
output_dir	Main output folder. If the script's folder contains FIES-inputs, FIES artifacts are written to FIES-outputs below this folder.

Value

The session, invisibly.

read_mics_tabulation	<i>Read one sheet from a MICS Plus tabulation plan</i>
----------------------	--

Description

Read one sheet from a MICS Plus tabulation plan

Usage

```
read_mics_tabulation(session, path_tab_excel, sheet)
```

Arguments

session	A session created by <code>mics_session()</code> .
path_tab_excel	Path to the Excel tabulation plan.
sheet	Sheet name or index.

Value

The parsed tabulation-plan context.

read_previous_mics_table

Read a previously generated worksheet for comparison

Description

Read a previously generated worksheet for comparison

Usage

```
read_previous_mics_table(path, sheet, skip = 3)
```

Arguments

path	Excel workbook path.
sheet	Sheet name or index.
skip	Number of header rows to skip.

Value

A cleaned table.

run_app

Run the micsPlusTableR Shiny application

Description

Starts the Shiny application bundled with micsPlusTableR.

Usage

```
run_app(
  host = "127.0.0.1",
  port = getOption("shiny.port"),
  launch.browser = interactive(),
  output_dir = NULL,
  ...
)
```

Arguments

host	Host interface passed to shiny::runApp().
port	Port passed to shiny::runApp(); NULL selects a free port.
launch.browser	Whether to open the application in a browser.
output_dir	Default directory for generated Excel workbooks. When NULL, the app creates micsPlusTableR-output in the project root detected by here::here().
...	Additional arguments passed to shiny::runApp().

Value

Invisibly returns the value from shiny::runApp().

survey_choices

Maintain survey and wave choices

Description

The plain UTF-8 CSV file ‘inst/extdata/survey_choices.csv’ in the package source controls the application’s survey and wave selectors. Edit this source file directly before committing to GitHub or building the package.

Columns

Each row represents one available country, period, and wave combination. All four columns are read as character strings:

country Full country name, for example Jamaica.

country_code Exactly three uppercase letters, for example JAM.

period Survey period as text, for example 2023–24.

wave Wave label, for example Wave 1.

Keep the column names. The file must have at least one row, with no empty fields or duplicate country/period/wave combinations. All waves of the same country and period must use the same country code. Leading and trailing whitespace is removed. Row order determines selector order and defaults.

Editing and installation

Use a text editor or a spreadsheet editor that preserves the period and wave columns as text. Save as UTF-8 CSV. From the package root, run `source("run_dev.R")` to load and launch the development source. Stop and restart the app after saving changes.

The CSV requires no conversion to an R data object.

R installs ‘inst/extdata’ files into the package’s ‘extdata’ directory. For installed packages, reinstall the updated source and restart the app. Direct edits to an installed copy are replaced by re-installation or upgrades. The selected code, full period, and wave also contribute to output filenames and long-format metadata.

See Also

[run_app, micsPlusTableR-package](#)

Examples

```
path <- system.file("extdata", "survey_choices.csv", package = "micsPlusTableR")
choices <- read.csv(path, colClasses = "character", fileEncoding = "UTF-8-BOM")
choices
```

tabulate_mics_table	<i>Tabulate a MICS Plus table outside the application</i>
---------------------	---

Description

Tabulate a MICS Plus table outside the application

Usage

```
tabulate_mics_table(session, path_tab_excel = NULL, sheet = NULL)
```

Arguments

session	A prepared tabulation session.
path_tab_excel	Optional tabulation-plan path. Supply with sheet to read a new table before tabulating.
sheet	Optional sheet name or index.

Value

A long-format data frame containing one row per calculated cell.

validate_sdmx_compatible	<i>Validate SDMX-compatible MICS long-format data</i>
--------------------------	---

Description

Checks the complete 35-variable contract and the unique observation key: SURVEY_ID, TABLE_ID, row_order, column_order, and TIME_PERIOD.

Usage

```
validate_sdmx_compatible(data)
```

Arguments

data	A data frame produced by as_sdmx_compatible() .
------	---

Value

data, invisibly.

write_mics_footnotes	<i>Add standard footnotes to a tabulation workbook</i>
----------------------	--

Description

Add standard footnotes to a tabulation workbook

Usage

```
write_mics_footnotes(session, destination, sheet = NULL, table = NULL)
```

Arguments

session	A tabulation session.
destination	Existing destination workbook.
sheet	Sheet name or index; defaults to the current sheet.
table	Cell results; defaults to the latest result.

Value

The destination path, invisibly.

write_mics_table	<i>Write cell results to a tabulation workbook</i>
------------------	--

Description

Write cell results to a tabulation workbook

Usage

```
write_mics_table(
  session,
  destination,
  sheet = NULL,
  table = NULL,
  formatted = FALSE,
  drop_n_unw = FALSE
)
```

Arguments

session	A tabulation session with a loaded plan.
destination	Existing destination workbook.
sheet	Sheet name or index; defaults to the current sheet.
table	Cell results; defaults to the latest result.
formatted	Whether to write formatted display values.
drop_n_unw	Whether to omit unweighted-count columns when formatting.

Value

The destination path, invisibly.

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